

Versatile and Amplified Biosensing through Enzymatic Cascade Reaction by Coupling Alkaline Phosphatase *in Situ* Generation of Photoresponsive Nanozyme

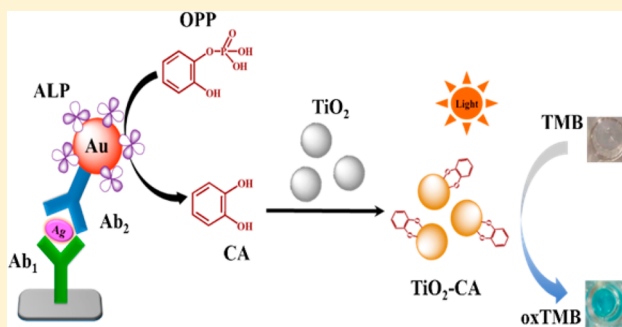
Lu-Yi Jin,[†] Yu-Ming Dong,[†] Xiu-Ming Wu,[†] Gen-Xia Cao,[†] and Guang-Li Wang^{*,†,‡}

[†]The Key Laboratory of Food Colloids and Biotechnology, Ministry of Education, School of Chemical and Material Engineering, Jiangnan University, Wuxi 214122, Jiangsu, China

[‡]State Key Laboratory of Analytical Chemistry for Life Science, Nanjing University, Nanjing 210093, Jiangsu, China

S Supporting Information

ABSTRACT: The alkaline phosphatase (ALP) biocatalysis followed by the *in situ* enzymatic generation of a visible light responsive nanozyme is coupled to elucidate a novel amplification strategy by enzymatic cascade reaction for versatile biosensing. The enzymatic hydrolysis of *o*-phosphonoxyphehol (OPP) to catechol (CA) by ALP is allowed to coordinate on the surface of TiO₂ nanoparticles (NPs) due to the specificity and high affinity of enediol ligands to Ti(IV). Upon the stimuli by CA generated from ALP, the inert TiO₂ NPs is activated, which demonstrates highly efficient oxidase mimicking activity for catalyzing the oxidation of the typical substrate of 3,3',5,5'-tetramethylbenzidine (TMB) under visible light ($\lambda \geq 400$ nm) irradiation utilizing dissolved oxygen as an electron acceptor. On the basis of the cascade reaction of ALP and the nanozyme of CA coordinated TiO₂ (TiO₂-CA) NPs, we design exquisitely colorimetric biosensors for probing ALP activity and its inhibitor of 2, 4-dichlorophenoxyacetic acid (2,4-DA). Quantitative probing of ALP activity in a wide linear range from 0.01 to 150 U/L with the detection limit of 0.002 U/L is realized, which endows the methodology with sufficiently high sensitivity for potentially practical applications in real samples of human serum (ALP level of 40–190 U/L for adults). In addition, a novel immunoassay protocol by taking mouse IgG as an example is validated using the ALP/nanozyme cascade amplification reaction as the signal transducer. A low detection limit of 2.0 pg/mL is attained for mouse IgG, which is 4500-fold lower than that of the standard enzyme-linked immuno-sorbent assay (ELISA) kit. Although only mouse IgG is used as a proof-of-concept in our experiment, we believe that this approach is generalizable to be readily extended to other ELISA systems. This methodology opens a new horizon for amplified and versatile biosensing including probing ALP activity and following ALP-based ELISA immunoassays.



Alkaline phosphatase (ALP) exists in a variety of tissues and plays significant roles in regulating intracellular processes involved in cell cycle, growth, apoptosis, and signal transduction pathways in living organisms.^{1,2} It is also an important biomarker connecting to several diseases, such as adynamic bone disease, hepatitis, prostatic cancer, and so on.^{3,4} Importantly, ALP is one of the most widely used labeling tracer in enzyme-linked immuno-sorbent assay (ELISA) due to its high catalytic activity, good stability, low cost, and broad substrate specificity.^{5–7} In ELISA, the specific interaction between an antigen and an antibody is tethered to certain detectable label, for example, enzyme labels, to produce a measurable signal. Nowadays, ELISA-based colorimetric immunoassays, with the advantages of simplicity, cost effectiveness, and rapid/direct readout by the naked eye, have grown as a powerful technique with applications in various fields including environmental monitoring, food safety, and clinical diagnosis.^{8–12} Therefore, highly sensitive detection of ALP activity is needed in various diagnostic applications. The majorities of the currently developed methods for ALP activity

detection^{13–15} or ALP-based immunoassays^{5,16} merely rely on the catalytic reaction of ALP for signal readout, which are simple but suffer from poor sensitivity due to limited signal amplification strategy. Thus, it is still highly desirable to construct innovative and powerful systems for ALP activity detection.

With the development of nanoscience and nanotechnology, nanomaterial-based signal amplification has aroused increasing attention because of its ultrasensitivity for analytes with additional benefits of rapidness and easy miniaturization. Enzyme mimetics based on nanomaterials, termed nanozymes, are fascinating due to their prominent merits including highly efficient activity, facile preparation and purification, robustness in harsh conditions, tailored enzymatic activities, and design flexibility.^{17,18} Most the currently developed nanozymes are

Received: July 6, 2015

Accepted: September 29, 2015



peroxidase mimetics, which catalyze the oxidation of the typical substrates using H_2O_2 as an oxidant.^{17,18} Especially, the coupling of a natural enzyme with the enzymatically generated nanozyme of peroxidase mimetics seems to be an exquisite tactic in biosensing due to dual-enzyme cascade amplification. Tang's group¹⁹ implemented ALP-biocatalyzed hydrolysis of ascorbic acid 2-phosphate (AAP) to produce ascorbic acid (AA), which reduced H_2PdCl_6 and generated palladium nanostructures (Pd NS) on the surface of Au nanoparticles (NPs). The as produced Pd NS exhibited highly peroxidase-like activity to catalyze the oxidation of 3,3',5,5'-tetramethylbenzidine (TMB) using H_2O_2 as an oxidant, which enabled a dual amplified immunoassay protocol. Despite the successfully *in situ* generation of the nanozyme (Pd NS) with peroxidase mimicking activity through the ALP biocatalysis, the obstacle in applications of this peroxidase mimetics (Pd NS) arise from the harsh reaction conditions (a high concentration (7.2 mol/L) of H_2O_2 is required), which severely cripples the stability of natural biomolecules.

Recently, our group has attempted to explore novel nanozymes that can work well under mild conditions, especially those not in need of destructive H_2O_2 . Luckily, we have developed a set of light regulated nanozymes, such as AgX (X = Cl, Br, I) NPs,²⁰ graphene oxide,²¹ and gold nanoclusters,²² whose oxidase-like activities are light triggered in the absence of H_2O_2 . These light responsive nanozymes utilizing dissolved oxygen as an electron acceptor endow them good biocompatibility as a transducer in biosensing applications. Unfortunately, like most of the currently developed nanozymes possessing peroxidase-like activity,^{17,18} all these light responsive nanozymes are ready made nanomaterials, which are difficult to be coupled with other natural enzymes for cascade signal amplifications.

Herein, we introduce a new concept to produce a phototriggered nanozyme through the ALP-based *in situ* biocatalytic process. The dephosphorylation process of *o*-phosphonoxyphephenol (OPP) by ALP generated the product of catechol (CA) containing enediol-ligand, which had specificity and large affinity for the uncoordinated Ti atoms on the surface of TiO_2 NPs and formed the CA–Ti (IV) charge transfer (CT) complex.²³ Upon visible light triggering, unlike the inertness of the native TiO_2 , the TiO_2 NPs with the surface coordinated CA–Ti (IV) CT complex demonstrated oxidase-like activity under visible light ($\lambda \geq 400$ nm) stimulation. Also, this photoactive nanozyme possessed advantages of high catalytic activity, controllable and tailored activities by light irradiation, and mild reaction conditions without destructive H_2O_2 . Therefore, a cascade reaction was coupled for probing ALP activity with high sensitivity: ALP catalyzed the generation of CA and, in turn, activated a second nanozyme (TiO_2 –CA), resulting in efficient signal amplification. As a proof of concept, ALP-based colorimetric immunoassay using mouse IgG as a model analyte was constructed based on the signal-transduction tag by coupling ALP and *in situ* generated nanozyme of TiO_2 –CA, which was also proved to be highly sensitive. The high sensitivity of the protocols for probing ALP activity and the ALP-based immunoassay was resulted from the exquisite consolidation of the cascade reaction of ALP and the enzymatically generated nanozyme (TiO_2 –CA). To the best of our knowledge, this is the first example of *in situ* generated photoactive nanozyme by a naturally enzymatic process, which may expand the scope of biosensing though biocatalytically generated photoactive nanozymes.

EXPERIMENTAL SECTION

Reagents. Superoxide dismutase (from bovine liver), horseradish peroxidase, and glucose oxidase were purchased from Sigma-Aldrich (St. Louis, MO). Alkaline phosphatase (ALP) and thrombin were purchased from Shanghai Yuanye Biological Technology Co., Ltd. (Shanghai, China). Mouse IgG, polyclonal goat antimouse IgG, human IgG, and goat IgG were obtained from Boster Biological Technology, Co., Ltd. (Wuhan, China). Rabbit IgG was purchased from Shfeng Biological Technology Co., Ltd. (Shanghai, China). Tetra-*n*-butyl titanate, 3,3',5,5'-tetramethylbenzidine, catechol, ascorbic acid, salicylic acid, phenol, phosphorus pentoxide, methanol, *t*-butanol, isopropanol, ethylene diamine tetraacetic acid, KI, bovine serum albumin, trypsin, lysozyme, and Tween 20 were all obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China).

The dilution and usage of ALP should be with 0.01 mol/L Tris-HCl (pH 9.0) containing 5 mmol/L MgCl_2 and 0.2 mmol/L ZnCl_2 . A 0.01 mmol/L phosphate-buffered saline (PBS, pH 7.4) containing 0.15 mol/L NaCl was used for the preparation of the antigen (Ag) and antibody (Ab) stock solutions. In the immunoassay, the washing buffer solution was 0.01 mol/L PBS (pH 7.4, 0.15 mol/L NaCl) containing 0.05% Tween 20. The blocking buffer was 0.01 mol/L PBS (pH 7.4) containing 3% (w/v) bovine serum albumin (BSA).

Apparatus. High-resolution transmission electron microscopy (HRTEM) images were performed using a JEOL JEM-2100 transmission electron microscope (Hitachi, Japan). Scanning electron microscopy (SEM) images were captured on a Hitachi S-4800 high-resolution scanning electron microscope (Hitachi, Japan). The X-ray powder diffraction (XRD) pattern was obtained by an X'Pert Philips materials research diffractometer using $\text{Cu K}\alpha$ radiation and employing a scanning speed of $0.02^\circ \text{ s}^{-1}$ in the 2θ range of 20 to 75° . Fourier transform-infrared (FT-IR) spectra were recorded on a Nicolet FT-IR 6700 spectrometer (Thermo Fisher). ^1H -NMR spectra were recorded on an AVANCE III (400 MHz) instrument at 298 K using standard Bruker software (Bruker, Switzerland). UV–vis absorption spectroscopic measurements were carried out using a TU-1901 spectrophotometer (Beijing Purkinje General Instrument Co., Ltd., China). The detection of ALP and the immunoassay by the absorption method were conducted on 96 well plates by an enzyme labeled meter (SpectraMax M5). Photoelectrochemistry and linear potential scan measurements were conducted on a CHI 800C electrochemical workstation using a three-electrode system: TiO_2 or TiO_2 –CA modified ITO electrode was employed as the working electrode; and a Pt wire and a saturated Ag/AgCl were used as the counter and the reference electrode, respectively.

Hydrothermal Synthesis of TiO_2 NPs. TiO_2 NPs were synthesized according to a modified procedure.²⁴ First, 4.5 mL of tetra-*n*-butyl titanate was added into 75 mL of distilled water dropwise and then the mixture was stirred for 2 h at room temperature. Subsequently, the solution was transferred to a 100 mL Teflon-lined stainless autoclave and heated at 433 K for 24 h. The acquired products were filtered and washed with distilled water several times and dried at 333 K for 12 h.

Catalytic Oxidation of TMB by TiO_2 –CA under Visible Light ($\lambda \geq 400$ nm) Irradiation. The as obtained TiO_2 NPs were redispersed in water by ultrasonication. TiO_2 –CA NPs were acquired by mixing 250 μL of catechol (CA, 1.0×10^{-4} mol/L) and 1 mL of TiO_2 NPs solution (1.5 mg/mL) for a

moment and then centrifuged and redispersed in 1 mL of ultrapure water. To assess the catalytic activity of the as-prepared TiO_2 -CA NPs, 100 μL of TiO_2 -CA NPs in a reaction volume of 5 mL of acetate buffer solution (40 mmol/L, pH 4.0) with 500 $\mu\text{mol/L}$ TMB as the substrate were irradiated with a 300 W Xe lamp equipped with an ultraviolet cutoff filter ($\lambda \geq 400$ nm).

The reaction kinetics for the catalytic oxidation of TMB was studied by monitoring the absorption intensity at 652 nm (for oxidized TMB, oxTMB) with a 1 min interval. Catalytic experiments were carried out at 30 $^\circ\text{C}$ using 100 μL of TiO_2 -CA NPs in 5 mL of acetate buffer solution (40 mmol/L, pH 4.0) with varied concentrations of TMB under visible light irradiation.

Preparation of *o*-Phosphonoxyphenol (OPP). OPP was synthesized based on the method given by Tago et al.²⁵ CA (1.24 g, 11.25 mmol) was warmed at a temperature slightly higher than its melting point (110 $^\circ\text{C}$), and diphosphorus pentoxide (1.06 g, 7.5 mmol) was slowly added under vigorous stirring within 2 h. As the reaction proceeded, the mixture became a brownish viscous liquid. After cooling to room temperature, the mixture was dissolved in dichloromethane (300 mL) containing a few drops of methanol, followed by concentration. After cooling to room temperature, the precipitate was collected by filtration and dried in vacuum to give OPP as a white powder.

ALP Detection. The ALP activity assay was conducted on 96 well plates based on its enzymatic hydrolysis of OPP to generate CA for coordination with TiO_2 NPs to form a photoresponsive oxidase mimetic. First, 20 μL of ALP was mixed with the substrate of 50 μL 20 $\mu\text{mol/L}$ of OPP for an incubation time of 1 h at 37 $^\circ\text{C}$ in pH 9.0 Tris-HCl buffer solution. Following that, 10 μL of 1.5 mg/mL TiO_2 NPs, 100 μL of pH 4.0 acetate buffer solution, and 20 μL of 5 mmol/L TMB were added in the above solution and illuminated for 10 min at 30 $^\circ\text{C}$ under visible-light ($\lambda \geq 400$ nm). Finally, the absorbance of the oxTMB at 652 nm was measured by an enzyme labeled meter.

Immunoassay for Mouse IgG. Au NPs were used as nanocarriers for the co-immobilization of the secondary antibody (polyclonal goat antimouse IgG, Ab₂) and ALP to prepare Au NP/Ab₂/ALP bioconjugates.²⁶ The bioconjugates were freshly prepared by gently mixing the Ab₂ (20 μL , 0.5 mg/mL), ALP (40 μL , 0.5 mg/mL), and 1.0 mL of the Au NPs' solution for 2 h. After blocking of Au NPs by 200 μL of 1% BSA solution for 30 min at room temperature and centrifuging the mixture at 15 000 rpm for 30 min, the oiled drop was washed by the washing buffer, centrifuged again, and finally resuspended in 200 μL of 1% BSA.

To conduct the immunoassay, 25 μL of 0.5 mg/mL polyclonal goat antimouse IgG (first antibody, Ab₁) was first spread onto the 96 well plates at 4 $^\circ\text{C}$ overnight followed by rinsing with the washing buffer. The wells were then blocked with 25 μL of the blocking solution for 2 h at 4 $^\circ\text{C}$ to block nonspecific binding sites and washed with the washing buffer thoroughly. Then, 25 μL of mouse IgG (antigen, Ag) with different concentrations were dropped onto the Ab₁ modified wells for an incubation time of 60 min at 37 $^\circ\text{C}$ followed by washing. After the immunoreaction between the Ab₁ and the Ag, the wells were allowed for labeling by an additional incubation with 25 μL of Au NP/Ab₂/ALP bioconjugates' solution for 60 min and again washed thoroughly. After that, the plates were allowed for an incubation in 0.1 mol/L Tris-

HCl buffer (pH 9.0) containing 20 μL of 5 $\mu\text{mol/L}$ OPP at 37 $^\circ\text{C}$ for 60 min. Finally, 10 μL of 1.5 mg/mL TiO_2 , 100 μL of acetate buffer (pH 4.0), and 20 μL of 5 mmol/L TMB were added into the 96 well plates and illuminated for 10 min at 30 $^\circ\text{C}$ under visible light ($\lambda \geq 400$ nm) irradiation. Finally, the absorbance of oxTMB at 652 nm was measured.

RESULTS AND DISCUSSION

Characterization of CA Coordinated TiO_2 NPs (TiO_2 -CA) and Investigation of Their Photo Triggered Oxidase-Like Activity. The SEM and TEM images (Figure S1, Supporting Information) show the morphology and size distribution of the TiO_2 NPs synthesized according to a previously modified report,²⁴ which indicate that they are spherical in shape with an average size of about 3–5 nm. The XRD pattern confirms the tetragonal anatase structure of TiO_2 (JCPDS no. 21-1272). TiO_2 is a well-known wide band gap semiconductor, whose absorption is in the ultraviolet range. After the addition of catechol (CA) to the native TiO_2 NPs, the white colored TiO_2 NPs become yellow and a red shift in their absorption threshold into the visible range is observed (Figure 1A). The varied optical properties of TiO_2 NPs after mixing

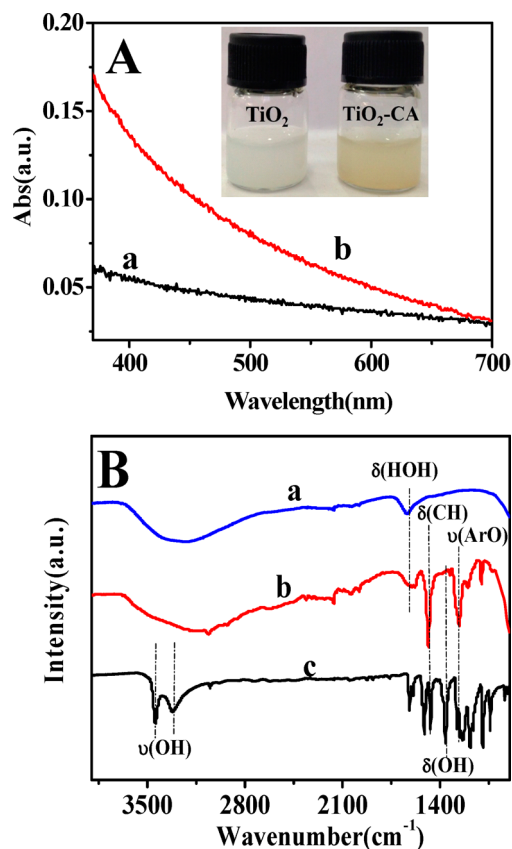


Figure 1. (A) Absorption spectra of native (a) and CA modified (b) TiO_2 NPs. Inset: the corresponding photographs of the above samples. (B) FT-IR spectra of (a) TiO_2 , (b) TiO_2 -CA, and (c) CA.

with CA results from the formation of a charge transfer (CT) complex on the surface of TiO_2 NPs.²³ When the size of NPs is under a certain regime, an increased fraction of the atoms that constitute the NPs are located at the surface, which significantly alter their surface properties. Though the surface Ti atoms are hexacoordinated (octahedral) in bulk TiO_2 , they are

pentacoordinated (square pyramidal) when the size of TiO_2 NPs become smaller than 20 nm.²³ Because of surface effects, molecules with enediol-ligands have a specific and large affinity for the undercoordinated surface Ti atoms on TiO_2 NPs, restoring the Ti atoms to the octahedral coordination and forming the enediol ligand–Ti(IV) CT complex.²³ The formation of the CT complex changes the electronic properties of TiO_2 NPs as reflected by the absorption spectrum due to the electron and hole confinement effect. FT-IR spectra confirm the formation the CA–Ti(IV) complex on the surface of TiO_2 NPs (Figure 1B). The stretching (3440 and 3320 cm^{-1}) and bending (1350 cm^{-1}) vibrations of –OH groups of CA disappear upon mixing with TiO_2 NPs, indicating its coordination with TiO_2 NPs through enediol groups. While the C–H bend at 1480 cm^{-1} and the aryl oxygen stretch at 1260 cm^{-1} of CA appear for CA modified TiO_2 NPs, indicating the successful conjugation of CA on TiO_2 NPs. The coordination of CA on TiO_2 NPs was found to be instantaneous within several seconds, indicating that the formation of TiO_2 –CA proceeded easily and fast.

Using 3,3',5,5'-tetramethylbenzidine (TMB) as a typically benign chromogenic substrate of the natural enzymes including peroxidase or oxidase,^{27,28} we examined the enzyme-like catalytic activities of native TiO_2 NPs and TiO_2 –CA under different conditions (Figure 2). Apparently, TMB develops a blue color in the presence of visible light illuminated TiO_2 –CA, while no change occurs in the presence of native TiO_2 NPs (Figure 2) or TiO_2 –CA without visible light illumination (Figure S2, Supporting Information). This indicates that the originally inert TiO_2 NPs is photoactive under visible light irradiation ($\lambda \geq 400$ nm) after their surface coordination by

CA. The catalytic oxidation of TMB to its oxidation product (oxTMB) accounts for the above blue color development, which gives characteristic absorbance at 369 and 652 nm (Figure 2A) originating from its distinctive charge-transfer complex²⁹ of diamine and diamine (insert of Figure 2A). This catalytic oxidation reaction for TMB by the illuminated TiO_2 –CA is similar to that of natural enzymes, indicating its enzyme-like activity. As is known, natural peroxidase or oxidase as a biocatalyst can catalyze the oxidation of TMB to oxTMB with distinctive blue color using H_2O_2 or dissolved oxygen as an oxidant.^{30,31} Result also indicates that dissolved oxygen in solution plays an important role in the photo-stimulated catalytic oxidation of TMB by TiO_2 –CA (Figure S3, Supporting Information). As a result, we call the TiO_2 –CA as a phototriggered oxidase mimetic. The illuminated TiO_2 –CA also catalyzes the oxidation of another chromogenic substrate of oxidase of (3-ethylbenzothiazoline-6-sulfonic acid) (ABTS), producing the typical green color with characteristic absorption peak at 417 nm (Figure S4, Supporting Information), which indicates that the oxidase-like acidity of TiO_2 –CA is not restricted to TMB. The catalytic activity of TiO_2 NPs modified by other compounds also with the hydroxyl group, such as salicylic acid, ascorbic acid, or phenol, was also investigated. Salicylic acid coordinated TiO_2 NPs only shows very weak catalytic activity under visible light irradiation, while TiO_2 NPs modified by ascorbic acid or phenol does not express any photoinduced catalytic activity (Figure S5, Supporting Information), indicating that this nanozyme based on TiO_2 NPs is specific for CA containing enediol-ligands. It was found that the size of TiO_2 NPs influenced the phototriggered oxidase-like activity of the formed TiO_2 –CA. As the size of TiO_2 NPs decreases, the catalytic activity of the formed TiO_2 –CA increases (Table S1, Supporting Information). This can be explained that the surface sites available for the formation of the charge transfer complex decrease when the sizes of TiO_2 NPs increase.²³ The binding of enediol-ligands to TiO_2 NPs occurs only with smaller particle sizes ($D < 20$ nm).²³ Although Fe_2O_3 NPs were also found to coordinate with molecules having enediol-ligands,²³ we did not observe the similar photoinduced enzyme-like activity as that of TiO_2 –CA (data not shown). We speculate that the efficient separation of the photogenerated charges on the surface of TiO_2 –CA NPs³² enable their efficiently photoinduced enzyme-like activity.

Catalytic Characteristics and Mechanism of the Photoactivated Oxidase Mimetics of TiO_2 –CA NPs. The relative catalytic activity of TiO_2 –CA was, like natural enzymes, dependent on pH, temperature, and substrate concentration. The optimal pH and temperature of the catalytic activity of TiO_2 –CA is found at 4.0 and 30 °C, respectively (Figure S6, Supporting Information). With the concentrations of the substrate (TMB) increase, the as formed oxTMB increase accompanied by an increase in the absorption intensity (Figure S7A, Supporting Information). Furthermore, steady-state kinetics was used to study the oxidase-like activity of photostimulated TiO_2 –CA by monitoring the absorbance changes of oxTMB at 652 nm (Figure S7B, Supporting Information). The apparent kinetic parameters are calculated based on the function: $v = V_{\text{max}} \times [S]/(K_m + [S])$, where v is the initial velocity, V_{max} is the maximal reaction velocity, $[S]$ is the concentration of substrate, and K_m is the Michaelis constant which reflects the affinity of the enzyme for the substrate.³³ Typically, the apparent K_m and V_{max} value for the TiO_2 –CA with TMB as the substrate is 107 μM and 156.5 nM S^{-1} ,

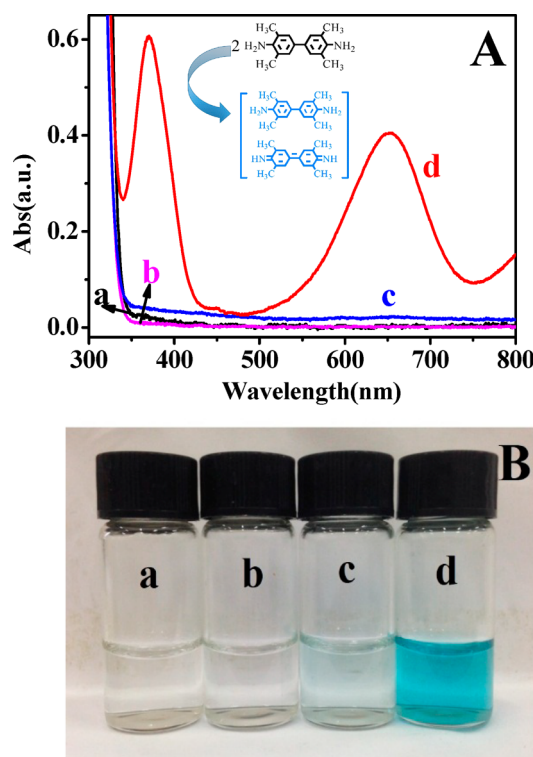


Figure 2. UV–vis absorption spectra (A) and the corresponding photographs (B) of different solutions under visible light illumination: (a) TMB, (b) TMB + CA, (c) TMB + TiO_2 NPs, (d) TMB + TiO_2 –CA.

respectively. The K_m value for the photoactivated TiO_2 -CA with TMB as the substrate is much lower than that of HRP ($K_m = 434 \mu\text{M}$),³⁴ suggesting that the TiO_2 -CA has higher affinity for TMB than natural HRP. In addition, this K_m value for the photoactivated TiO_2 -CA is also lower than the peroxidase mimetics of Pd NS ($K_m = 165 \mu\text{M}$) generated by the ALP-based *in situ* biocatalysis.¹⁹ However, unlike the widely studied peroxidase mimetics based on nanomaterials, the photoactivated TiO_2 -CA with oxidase-like activity possesses some advantages: (i) its activity can be easily regulated by switching the “on” and “off” state of light irradiation and (ii) the utilization of molecular oxygen instead of high amount of destructive H_2O_2 as an electron acceptor can ensure a greener and milder reaction condition, which makes it more appropriate as a signal transducer in biosensing applications.

To reveal the catalytic mechanism of photoactivated TiO_2 -CA, different quenchers (Figure 3A) were introduced into the catalytic reaction system to scavenge the possible reactive species, such as photo-generated holes (h^+), hydroxyl radicals ($\cdot\text{OH}$), superoxide anions ($\text{O}_2^{\cdot-}$), and H_2O_2 . The addition of isopropanol or *t*-butanol for scavenging $\cdot\text{OH}$ ³⁵ does not cause variation in the catalytic reaction. Nor the photo-stimulated catalytic activity is affected by catalase (CAT), an efficient enzyme for decomposing of H_2O_2 to water and oxygen,³⁶ indicating that H_2O_2 is not produced by illuminated TiO_2 -CA. On the contrary, the catalytic activity of TiO_2 -CA is apparently suppressed in the presence of ethylene diamine tetraacetic acid (EDTA) or KI, indicating that photogenerated holes (h^+)^{37,38} are one of the oxidative species. In addition, the absorbance of the oxTMB (652 nm) decreases obviously after the addition of superoxide dismutase (SOD), an antioxidant enzyme to effectively scavenge the $\text{O}_2^{\cdot-}$, showing that $\text{O}_2^{\cdot-}$ radicals are also the main reactive species in the catalytic reaction. On the basis of the above results, the photochemical processes of the TiO_2 -CA are proposed in Scheme 1. The coordination of CA on TiO_2 NPs makes the TiO_2 -CA absorb light in the visible regime ($\lambda \geq 400 \text{ nm}$). Upon visible light excitation, the CT complex of CA-Ti(IV) yields the instantaneous electron transfer from the donating ligands directly into the conduction band (CB) of TiO_2 NPs without transitioning through the excited state,^{23,32} leaving holes on CA to react with TMB directly. On the other hand, the photogenerated electrons on CB of TiO_2 NPs might be trapped by dissolved oxygen (as an electron acceptor), forming $\text{O}_2^{\cdot-}$ radicals. To validate the photoinduced charge transfer of TiO_2 -CA, the experiment of photoelectrochemistry was conducted. The rapid rise of the photocurrent of TiO_2 -CA upon illumination (Figure 3B) is indicative of the fast charge generation, separation, and transfer in the CT complex modified TiO_2 NPs that correlates with the surface chelating. The CB of the TiO_2 -CA estimated by cyclic voltammetry (CV) measurement³⁹ is -0.572 V (vs saturated Ag/AgCl) (Figure 3C), which is more negative than the reduction potential of oxygen (-0.046 V vs NHE⁴⁰). As a result, the excited electrons on the CB of TiO_2 can thermodynamically capture dissolved oxygen to generate reactive species of $\text{O}_2^{\cdot-}$ radicals.

Versatile Biosensors Based on the ALP/Nanozyme Cascade Reaction. The photoactive oxidase mimetic based on TiO_2 NPs was stimuli responsive to the presence of CA. As the concentrations of CA increase, the TiO_2 NPs-based mimicking activities are intensified obviously (Figure S8, Supporting Information). The observed oxidase mimicking activity of TiO_2 NPs is logarithmically proportional to the

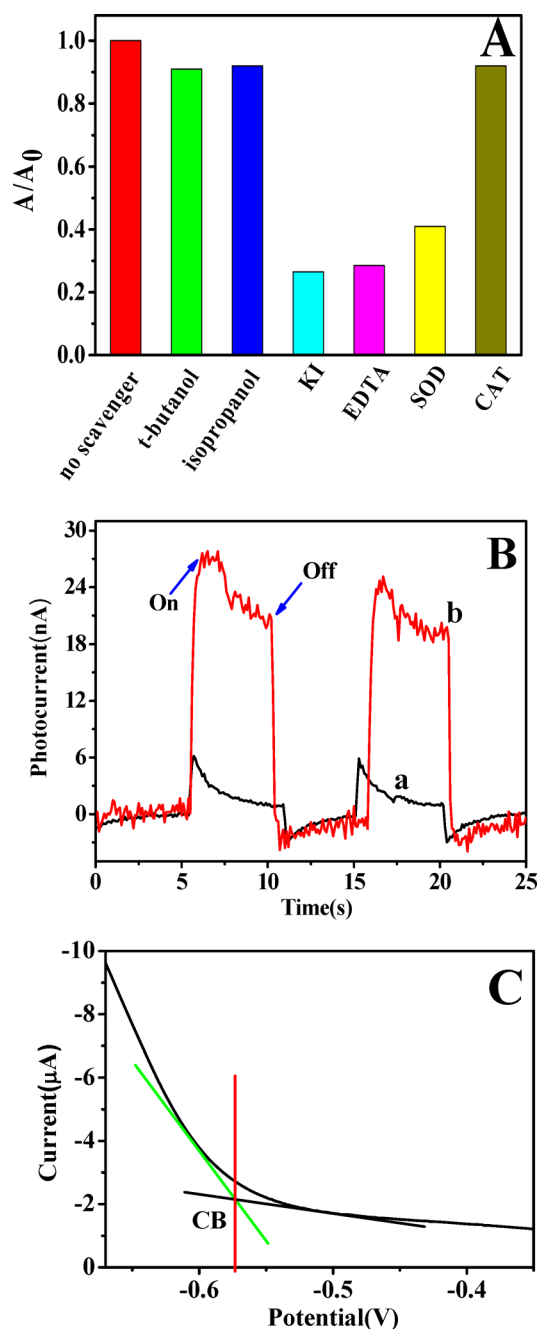
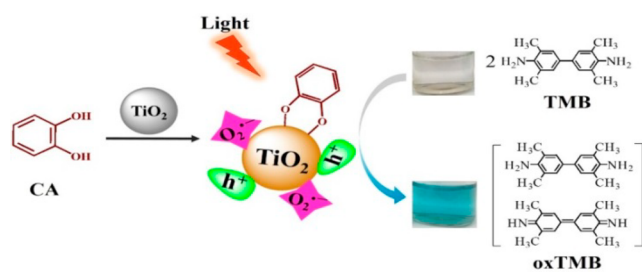


Figure 3. (A) Effect of different scavengers on the oxidation of TMB by illuminated TiO_2 -CA. (B) Photocurrent–time performances of TiO_2 (a) and TiO_2 -CA (b) modified electrodes at an applied potential of 0 V (vs saturated Ag/AgCl) in 0.1 mol/L Na_2SO_4 aqueous solution under visible light irradiation ($\lambda \geq 400 \text{ nm}$). (C) Cathodic linear potential scan for determining the CB edge of TiO_2 -CA specimens at 5 mV/s.

concentration of CA in the range of 0.05–500 nM with a detection limit of 0.01 nM ($S/N = 3$), indicating that this nanozyme is highly sensitive to CA. Triggering the enzyme mimicking activity of the TiO_2 NPs by CA stimuli enabled us to implement versatile bioassays. First, we probed the activity of ALP and its inhibitor of 2,4-dichlorophenoxyacetic acid (2,4-DA). ALP has wide substrate specificity and catalyzes the hydrolysis of a variety of orthophosphoric monoesters to yield phosphate and corresponding alcohols/phenols. *o*-Phosphoxyphenol (OPP) is one of the substrates of ALP, which can

Scheme 1. Proposed Mechanism of Catalytic Oxidation of TMB by Photo-Activated TiO_2 -CA



be converted to catechol (CA).⁴¹ OPP was synthesized by a reported procedure²⁵ and its formation was confirmed by FT-IR and ¹H-NMR (Figure S9, Supporting Information for detailed results). FT-IR spectrum of OPP has broad absorption bands ascribed to the phosphoric acid group at 1247 cm^{-1} ($\nu_{\text{P=O}}$ of $-\text{PO}_3\text{H}_2$), 1184 cm^{-1} ($\nu_{\text{P=O}}$ of $-\text{HPO}_3^-$), and 1092 cm^{-1} ($\nu_{\text{P=O}}$ of $-\text{PO}_3^{2-}$). The peak at 1463 cm^{-1} ($\delta_{\text{P-O-C}}$) and the shoulder at 1276 cm^{-1} ($\nu_{\text{P-O-C}}$) support the phosphonoxy structure. The four distinct chemical shifts (6.93, 6.88, 6.70, and 6.61) of ¹H in benzene ring indicate characteristics of monosubstituted CA after reaction of CA with phosphate. In a control experiment without the use of ALP, the oxidation of TMB in the presence of OPP alone was like that of native TiO_2 NPs, indicating that OPP did not contribute to the enzyme-like activity of TiO_2 NPs. Thus, the ALP activity was quantified by a cascade reaction with dual amplification strategies: first, ALP catalyzed the conversion of OPP into CA, and the *in situ* enzymatically generated CA then coordinated with TiO_2 NPs to form a visible light responsive oxidase mimetic with efficient capability for the catalytic oxidation of TMB to oxTMB. The colorimetric method for the detection of ALP manifests a linear range from 0.01 to 150 U/L (Figure 4) with a detection limit of 0.002 U/L. ALP can be detected by the naked eye even at 0.01 U/L (Figure 4A, insert image). The normal level of ALP in adult serum is 40–190 U/L.⁴² The established colorimetric assay based on the dual enzymes cascade amplification strategy endows the protocol with ultrahigh sensitivity that is potentially capable of practical detection of the ALP level in clinical analysis. This detection limit is a thousand- or hundred-fold lower than the conventional ALP detection method (4.3 U/L) or the recent report of ALP-based *in situ* generation of peroxidase mimetics of Pd NS for cascade amplification (0.22 U/L).¹⁹ In addition, this detection limit is also lower than other assays for ALP detection proposed in the literature^{13,15,43–48} (Table S2, Supporting Information). Moreover, the proposed method can achieve facile and rapid determination of ALP by the naked eye and no sophisticated instruments are required. Control experiments using BSA, trypsin, lysozyme, thrombin, and glucose oxidase (GO_x) demonstrate good selectivity of the method (Figure 4C).

The proposed method as an enzymatic toxin biosensor was also evaluated considering the importance of screening ALP inhibition by miscellaneous drugs, pesticides, medicines, and environmental pollutants.⁴⁹ Figure S10 in the Supporting Information displays the change of absorption spectra of the detection system in the presence of the inhibitor of 2,4-dichlorophenoxyacetic acid (2,4-DA). Because of the inhibition effect of 2,4-DA on ALP, the absorption of oxTMB markedly decreases than that without the inhibitor, demonstrating the feasibility of this protocol to sense ALP inhibition. The linear

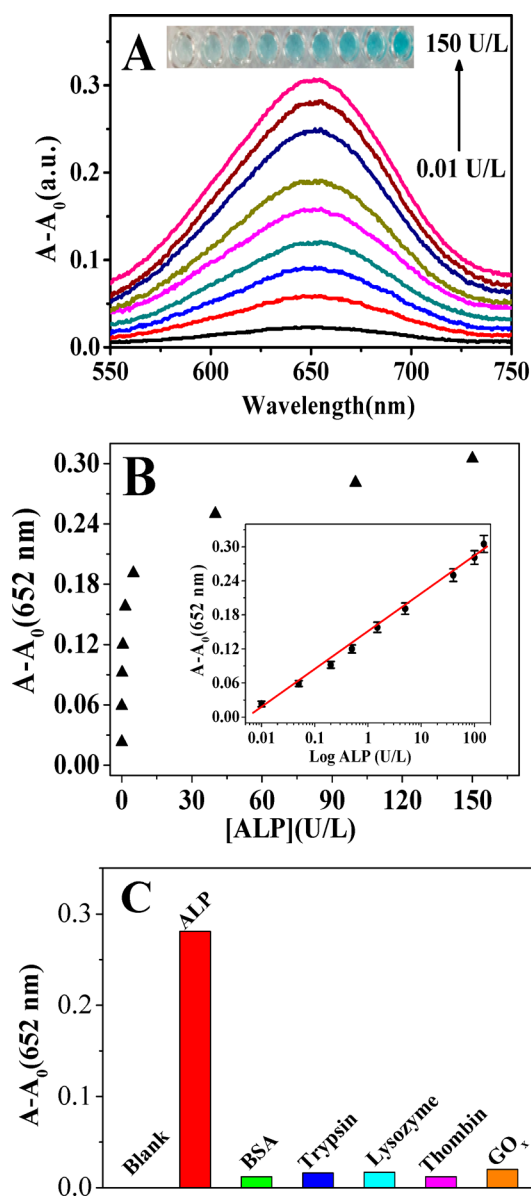
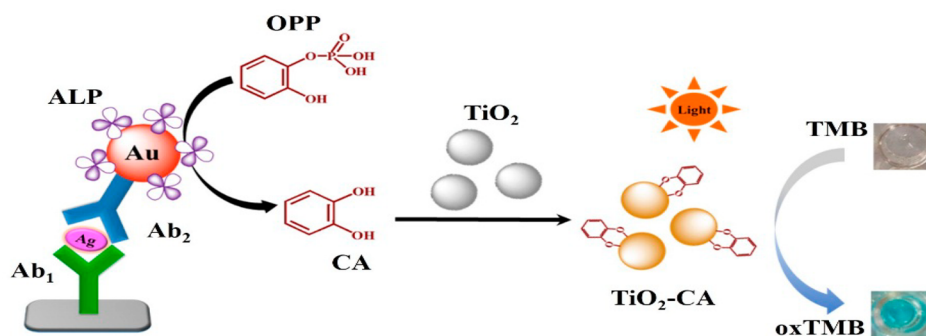


Figure 4. (A) UV-vis absorption spectra of the illuminated mixture of the TiO_2 NPs, OPP, and TMB in the presence of different concentrations of ALP. Inset: the corresponding photographs of the solutions. (B) The absorbance of oxTMB at 652 nm catalyzed by visible light illuminated TiO_2 NPs and OPP in the absence (defined as A_0) and presence (defined as A) of different concentrations of ALP. Inset: The linear calibration plots for ALP detection. Error bars represent the standard deviation of four repeated measurements. (C) Selectivity of the method for ALP detection. The concentrations of other substances are all 10 $\mu\text{g/mL}$ and the concentration of ALP is 0.05 $\mu\text{g/mL}$.

range for detecting 2,4-DA ranges from 0.1 to 100 $\mu\text{g/mL}$ with a detection limit of 0.02 $\mu\text{g/mL}$.

Considering the significance of ALP as a biocatalytic label widely used in immunoassays, the enzymatic cascade reaction combining ALP and the *in situ* generated nanozyme of TiO_2 -CA was applied in a colorimetric ELISA immunoassay by taking mouse IgG as a model (Scheme 2). ALP and mouse IgG secondary antibody (Ab_2) was co-immobilized on Au NPs for signaling the immunoreaction. The primary antibody (Ab_1) was immobilized on the microtiter plates to recognize the antigen (mouse IgG) and then conjugated with Ab_2 /ALP co-

Scheme 2. Proposed Immunodetection Process for Mouse IgG by Coupling the Cascade Reaction of ALP and the Enzymatically *In Situ* Generated Photoresponsive Nanozyme of TiO_2 –CA



immobilized Au NPs, forming a sandwich structure. After the hydrolysis of OPP to CA by the labeled ALP, TiO_2 NPs were added to form the nanozyme of TiO_2 –CA for further signal amplification. Figure S11A (Supporting Information) displays that the absorbance of oxTMB at 652 nm is associated with the concentrations of mouse IgG in the sandwich immunoassay format. The detection limit for mouse IgG is 2.0 pg/mL with a linear range from 0.01 to 100 ng/mL. The detection limit of the developed methodology for mouse IgG is close to 4500-fold lower than that of standard ELISA kit (LOD, 9.375 ng/mL).⁵⁰ Moreover, this detection limit is lower than other immunoassays for mouse IgG utilizing other different protocols^{51–58} (listed in Table S3, Supporting Information). This strategy for ALP-based immunoassay is very attractive because of its distinctive features: First, different from the conventional ELISA using multiple natural enzymes as labels, the oxidase mimetic of the nanostructured TiO_2 –CA is *in situ* generated by ALP mediated biocatalysis, which makes the detection more economic and facile. Second, this colorimetric strategy exhibits ultrahigh sensitivity by taking advantages of the ALP biocatalysis and the phototriggered nanozyme with efficient catalytic activity for cascade signal amplification. Third, the phototriggered nanozyme makes the detection more biocompatible than that of the peroxidase mimetics¹⁹ because its activity is dependent on dissolved oxygen not destructive H_2O_2 .

Responses from the 10-fold higher concentration of the interfering components of mouse IgG analogues of goat IgG, rabbit IgG, or human IgG is much lower than that of the target of mouse IgG (Figure S11B, Supporting Information), indicating acceptable selectivity of the method. The reproducibility of the developed immunoassay with cascade enzyme signal amplification were evaluated by calculating the variation coefficients (CVs, $n = 8$) of intra- and inter-assays. Experimental results indicated that the CVs of the assays using the same-batch immunosensing probes were between 2.1% and 5.9% in all cases. The obtained CV was about 6.3% by assaying 10 ng/mL mouse IgG (as an example) for eight times with different batches. The above results indicated that the developed methodology had good repeatability, revealing the possibility of batch detection.

The feasibility of the immunoassay system for potentially clinical applications was investigated using real samples of mouse serum. IgG levels in mouse serum is in the range of 5–13 mg/mL,⁵⁹ which is outside of the linear range of this assay. Therefore, the mouse serum was diluted 1:10⁶ in PBS solution to reduce IgG levels to an appropriate concentration. Absorbance changes of the detection system to the diluted

mouse serum sample (diluted 1:10⁶ in PBS) and its subsequent addition of different concentrations of mouse IgG are shown in Figure S12 (Supporting Information). The acceptable recoveries (determined by standard addition methods and calculated after deducting IgG response from the initially diluted mouse serum) ranging from 95.2% to 109% (shown in Table S4, Supporting Information) indicate that it is feasible to utilize the developed immunoassay for potentially practical applications.

CONCLUSIONS

In summary, an advanced signal-transduction protocol is proposed for constructing versatile colorimetric biosensors to detect ALP activity, ALP inhibition, and immunoassay of mouse IgG. By coupling the ALP-based hydrolysis of CA and the *in situ* formation of nanozyme of TiO_2 –CA NPs with efficiently phototriggered oxidase-like activity, a dual-enzyme cascade signal amplification strategy is elaborated. Especially, the enzymatically formed TiO_2 –CA NPs possess high oxidase-like activity upon light stimuli following a typical Michaelis–Menten kinetics and showing high affinity for the substrate (TMB). Importantly, this oxidase mimetic uses oxygen (not relies on destructive H_2O_2) as a mild and green electron acceptor, whose activity can be easily controlled by the switching “on/off” of light. Different from the ready-made nanozymes, this photoactive enzyme mimetic can be easily formed by an ALP-based biocatalytic process. Differentiated from the conventional ELISA using multiple natural enzymes as labels, the detectable signal adequately utilizes the high catalytic efficiency of ALP and the *in situ* generated oxidase mimetic of nanostructured TiO_2 –CA for cascade reactions, ensuring good sensitivity of the method. The developed method is facile, robust, sensitive, and needless of sophisticated instruments, which may easily be extended as a general protocol in high-throughput enzymatic assays and immunoassays.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.5b02728.

Characterization of TiO_2 NPs and OPP; evaluation of experimental conditions effects on the enzyme mimicking activity of TiO_2 –CA; feasibility of photoresponsive TiO_2 –CA strategy to ABTS; steady-state kinetic assay of TiO_2 –CA; application of the assay for detecting 2,4-DA; selectivity and recoveries of the immunoassay to mouse

IgG; and comparison of different assays for ALP or mouse IgG (PDF)

AUTHOR INFORMATION

Corresponding Author

*Fax: +86-510-85917763. E-mail: glwang@jiangnan.edu.cn.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grants 21275065 and 21005031), the Fundamental Research Funds for the Central Universities (Grant JUSRP51314B), and the Opening Foundation of the State Key Laboratory of Analytical Chemistry for Life Science of Nanjing University (Grant KLACLS1008).

REFERENCES

- (1) Coleman, J. E. *Annu. Rev. Biophys. Biomol. Struct.* **1992**, *21*, 441–483.
- (2) Johnson, L. N.; Lewis, R. J. *Chem. Rev.* **2001**, *101*, 2209–2242.
- (3) Colombatto, P.; Randone, A.; Civitico, G.; Gorin, J. M.; Dolci, L.; Medaina, N.; Oliveri, F.; Verme, G.; Marchiaro, G.; Pagni, R.; Karayiannis, P.; Thomas, H. C.; Hess, G.; Bonino, F.; Brunetto, M. R. *J. Viral Hepatitis* **1996**, *3*, 301–306.
- (4) Colombatto, P.; Randone, A.; Civitico, G.; Gorin, G. M.; Dolci, L.; Medaina, N.; Calleri, G.; Oliveri, F.; Baldi, M.; Tappero, G.; Volpes, R.; David, E.; Verme, G.; Smedile, A.; Bonino, F.; Brunetto, M. R. *J. Viral Hepat.* **1997**, *4*, 55–60.
- (5) Akanda, M. R.; Tamilavan, V.; Park, S.; Jo, K.; Hyun, M. H.; Yang, H. *Anal. Chem.* **2013**, *85*, 1631–1636.
- (6) Xianyu, Y.; Wang, Z.; Jiang, X. *ACS Nano* **2014**, *8*, 12741–12747.
- (7) Geraud, E.; Prevot, V.; Forano, C.; Mousty, C. *Chem. Commun.* **2008**, *13*, 1554–1556.
- (8) Lei, J.; Ju, H. *Chem. Soc. Rev.* **2012**, *41*, 2122–2134.
- (9) Zhang, B.; Tang, D.; Goryacheva, I.; Niessner, R.; Knopp, D. *Chem. - Eur. J.* **2013**, *19*, 2496–2503.
- (10) Lai, W. Q.; Tang, D. P.; Zhuang, J. Y.; Chen, G. N.; Yang, H. H. *Anal. Chem.* **2014**, *86*, 5061–5068.
- (11) Zhao, W. W.; Liu, Z.; Shan, S.; Zhang, W. W.; Wang, J.; Ma, Z. Y.; Xu, J. J.; Chen, H. Y. *Sci. Rep.* **2014**, *4*, 4426–4431.
- (12) Qu, W.; Liu, Y.; Liu, D.; Wang, Z.; Jiang, X. *Angew. Chem., Int. Ed.* **2011**, *50*, 3442–3445.
- (13) Qian, Z.; Chai, L.; Tang, C.; Huang, Y.; Chen, J.; Feng, H. *Anal. Chem.* **2015**, *87*, 2966–2973.
- (14) Zhang, L.; Zhao, J.; Duan, M.; Zhang, H.; Jiang, J.; Yu, R. *Anal. Chem.* **2013**, *85*, 3797–3801.
- (15) Wei, H.; Chen, C.; Han, B.; Wang, E. *Anal. Chem.* **2008**, *80*, 7051–7055.
- (16) Gao, Z.; Xu, M.; Hou, L.; Chen, G.; Tang, D. *Anal. Chem.* **2013**, *85*, 6945–6952.
- (17) Wei, H.; Wang, E. *Chem. Soc. Rev.* **2013**, *42*, 6060–6093.
- (18) Lin, Y. H.; Ren, J. S.; Qu, X. G. *Acc. Chem. Res.* **2014**, *47*, 1097–1105.
- (19) Gao, Z.; Hou, L.; Xu, M.; Tang, D. *Sci. Rep.* **2014**, *4*, 3966–3973.
- (20) Wang, G. L.; Xu, X. F.; Qiu, L.; Dong, Y. M.; Li, Z. J.; Zhang, C. *ACS Appl. Mater. Interfaces* **2014**, *6*, 6434–6442.
- (21) Wang, G. L.; Xu, X. F.; Wu, X. X.; Cao, G. X.; Dong, Y. M.; Li, Z. J. *J. Phys. Chem. C* **2014**, *118*, 28109–28117.
- (22) Wang, G. L.; Jin, L. Y.; Dong, Y. M.; Wu, X. M.; Li, Z. J. *Biosens. Bioelectron.* **2015**, *64*, 523–529.
- (23) Rajh, T.; Chen, L. X.; Lukas, K.; Liu, T.; Thurnauer, M. C.; Tiede, D. M. *J. Phys. Chem. B* **2002**, *106*, 10543–10552.
- (24) Wang, D.; Zhou, Z. H.; Yang, H.; Shen, K. B.; Huang, Y.; Shen, S. *J. Mater. Chem.* **2012**, *22*, 16306–16311.
- (25) Tago, T.; Kuwashiro, N.; Nishide, H. *Bull. Chem. Soc. Jpn.* **2007**, *80*, 1429–1434.
- (26) Zhao, W. W.; Ma, Z. Y.; Yan, D. Y.; Xu, J. J.; Chen, H. Y. *Anal. Chem.* **2012**, *84*, 10518–10521.
- (27) Gao, L. Z.; Zhuang, J.; Nie, L.; Zhang, J. B.; Zhang, Y.; Gu, N.; Wang, T. H.; Feng, J.; Yang, D. L.; Perrett, S.; Yan, X. *Nat. Nanotechnol.* **2007**, *2*, 577–583.
- (28) Xiong, Y. H.; Chen, S. H.; Ye, F. G.; Su, L. J.; Zhang, C.; Shen, S. F.; Zhao, S. L. *Chem. Commun.* **2015**, *51*, 4635–4638.
- (29) Jang, G. G.; Roper, D. K. *Anal. Chem.* **2011**, *83*, 1836–1842.
- (30) Le Goff, G. C.; Blum, L. J.; Marquette, C. A. *Anal. Chem.* **2011**, *83*, 3610–3615.
- (31) Asati, A.; Kaftanis, C.; Santra, S.; Perez, J. M. *Anal. Chem.* **2011**, *83*, 2547–2553.
- (32) Dimitrijevic, N. M.; Saponjic, Z. V.; Rabatic, B. M.; Rajh, T. *J. Am. Chem. Soc.* **2005**, *127*, 1344–1345.
- (33) Michaelis, L.; Menten, M. L. *Biochem. Z.* **1913**, *49*, 333–369.
- (34) Su, L.; Feng, J.; Zhou, X. M.; Ren, C. L.; Li, H. H.; Chen, X. G. *Anal. Chem.* **2012**, *84*, 5753–5758.
- (35) An, C. H.; Wang, J. Z.; Qin, C.; Jiang, W.; Wang, S. T.; Li, Y.; Zhang, Q. H. *J. Mater. Chem.* **2012**, *22*, 13153–13158.
- (36) Ammam, M.; Fransaer, J. *Sens. Actuators, B* **2011**, *160*, 1063–1069.
- (37) Cao, J.; Xu, B. Y.; Luo, B. D.; Lin, H. L.; Chen, S. F. *Appl. Surf. Sci.* **2011**, *257*, 7083–7089.
- (38) Feng, S.; Xu, H.; Liu, L.; Song, Y. H.; Li, H. M.; Xu, Y. G.; Xia, J. X.; Yin, S.; Yan, J. *Colloids Surf., A* **2012**, *410*, 23–30.
- (39) Das, A.; Li, T.; Nobusada, K.; Zeng, Q.; Rosi, N. L.; Jin, R. C. *J. Am. Chem. Soc.* **2012**, *134*, 20286–20289.
- (40) Chen, C.; Ma, W.; Zhao, J. C. *Chem. Soc. Rev.* **2010**, *39*, 4206–4219.
- (41) Szydłowska, D.; Campàs, M.; Marty, J. L.; Trojanowicz, M. *Sens. Actuators, B* **2006**, *113*, 787–796.
- (42) Hausamen, T. U.; Helger, R.; Rick, W.; Gross, W. *Clin. Chim. Acta* **1967**, *15*, 241–245.
- (43) Malashikhina, N.; Garai-Ibabe, G.; Pavlov, V. *Anal. Chem.* **2013**, *85*, 6866–6870.
- (44) Liang, J.; Kwok, R. T. K.; Shi, H. B.; Tang, B. Z.; Liu, B. *ACS Appl. Mater. Interfaces* **2013**, *5*, 8784–8789.
- (45) Ruan, C. M.; Li, Y. B. *Talanta* **2001**, *54*, 1095–1103.
- (46) Ruan, C.; Wang, W.; Gu, B. *Anal. Chem.* **2006**, *78*, 3379–3384.
- (47) Hayata, A.; Bulbul, G.; Andreescu, S. *Biosens. Bioelectron.* **2014**, *56*, 334–339.
- (48) Zhou, C. H.; Zhao, J. Y.; Pang, D. W.; Zhang, Z. L. *Anal. Chem.* **2014**, *86*, 2752–2759.
- (49) Millán, J. L. *Mammalian Alkaline Phosphatases. From Biology to Applications in Medicine and Biotechnology*; Wiley-VCH Verlag GmbH & Co: Weinheim, Germany, 2006.
- (50) Zhao, W. W.; Ma, Z. Y.; Yu, P. P.; Dong, X. Y.; Xu, J. J.; Chen, H. Y. *Anal. Chem.* **2012**, *84*, 917–923.
- (51) Dong, H.; Li, C. M.; Zhang, Y. F.; Cao, X. D.; Gan, Y. *Lab Chip* **2007**, *7*, 1752–1758.
- (52) Do, J.; Ahn, C. H. *Lab Chip* **2008**, *8*, 542–549.
- (53) Aziz, M. A.; Patra, S.; Yang, H. *Chem. Commun.* **2008**, *38*, 4607–4609.
- (54) Wang, G. L.; Yu, P. P.; Xu, J. J.; Chen, H. Y. *J. Phys. Chem. C* **2009**, *113*, 11142–11148.
- (55) Preechaworapun, A.; Ivandini, T. A.; Suzuki, A.; Fujishima, A.; Chailapakul, O.; Einaga, Y. *Anal. Chem.* **2008**, *80*, 2077–2083.
- (56) Avseenko, N. V.; Morozova, T. Y.; Ataulakhov, F. I.; Morozov, V. N. *Anal. Chem.* **2001**, *73*, 6047–6052.
- (57) Chen, Y. Y.; Cheng, H. W.; Tram, K.; Zhang, S. F.; Zhao, Y. H.; Han, L. Y.; Chen, Z. P.; Huan, S. Y. *Analyst* **2013**, *138*, 2624–2631.
- (58) Zhou, H.; Yue, H.; Zhou, Y. L.; Wang, L.; Fu, Z. F. *Sens. Actuators, B* **2015**, *209*, 744–750.
- (59) Penn, M. A.; Drake, D. M.; Driskell, J. D. *Anal. Chem.* **2013**, *85*, 8609–8617.